Exam 4 Review

Greatest hits of the semester

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DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

How to study this deck

- Use the weekly study guides as a starting point.
- Focus on the **mechanism**, the state it manages, and the tradeoffs it creates.
- Be ready to compare different **policies** and explain why one exists instead of another.
- If you can explain the idea in your own words, you are usually in good shape.

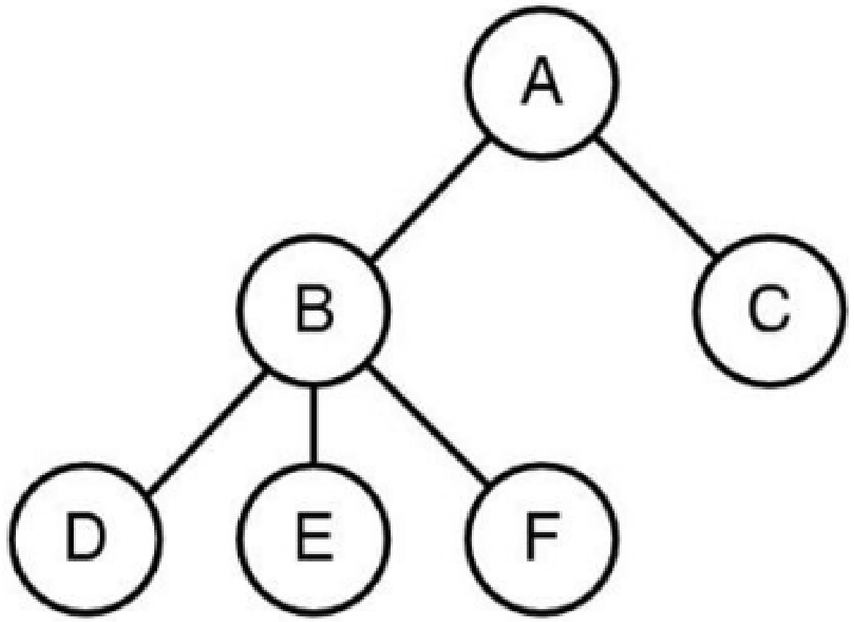
Exam overview

- 20-25 questions, mostly fill-in-the-blank, calculation, and short answer
- 110 minutes
- handwritten
- closed book, no internet
- one handwritten note sheet, both sides
- bring a dedicated calculator; phones do not count
- roughly 25% new material

Processes and Scheduling

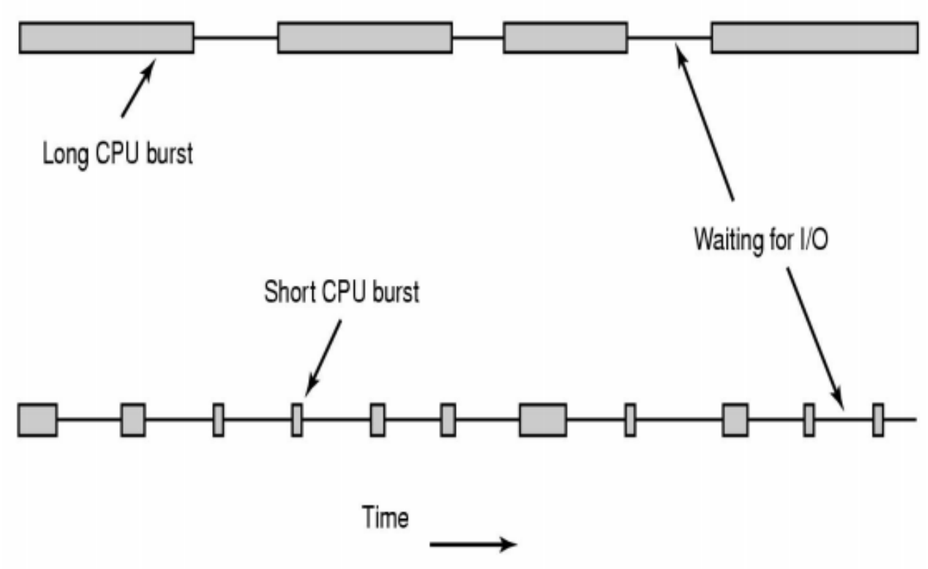
What runs, when it runs, and why the scheduler makes the choice

Processes



- A program is a file on disk; a process is the running instance.
- `fork()` clones the current process, `exec()` replaces its code, and `wait()` reaps children.
- The OS manages processes, and the shell is another process that launches them.
- On questions, draw the process tree and the ready/running/blocked/terminated state diagram first.
- Then reason about what state changed and why.

Scheduling

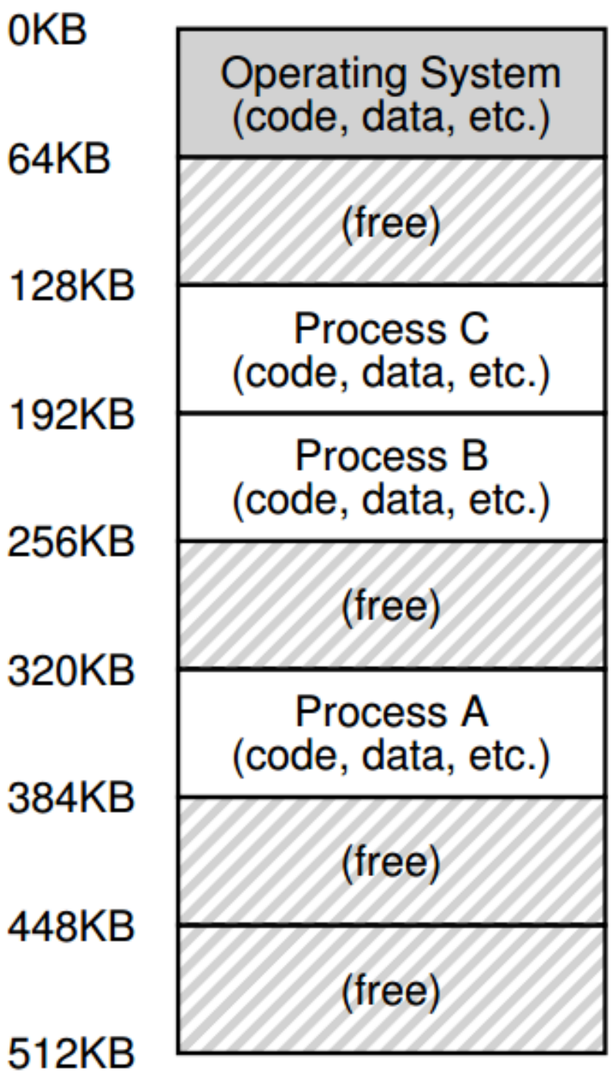


- Know the scheduling metrics: turnaround, response, utilization, and fairness.
- FCFS, SJF/STCF, and RR each optimize different goals and fail in different ways.
- MLFQ tries to favor short or interactive jobs while still giving CPU-bound jobs time.
- Be ready to simulate the next scheduling decision and explain the tradeoff being made.

Virtual Memory

How the machine gives each process its own illusion of memory

Address translation



- Think of translation as something that happens when code calls `read_byte(vaddr)` or `write_byte(vaddr, value)`.
- Virtual memory gives each process the illusion that it has its own private memory.
- Base-and-bounds, segmentation, paging, TLBs, and swapping are different **mechanisms** for implementing that illusion.
- Hardware does the translation; the OS sets up and maintains the state.
- Locality matters because extra memory references add up quickly.

Memory questions to be able to answer

- Translate a simple address by hand.
- Say what happens on a TLB miss.
- Explain why page tables can be hierarchical.
- Describe what swapping buys you and what it costs.
- Explain how free-space structures and fragmentation show up in memory management.

Concurrency

Why shared state makes everything interesting

Threads and locks

```c
#include <stdio.h>
#include <assert.h>
#include <pthread.h>

void *mythread(void *arg) {
    printf("%s\n", (char *) arg);
    return NULL;
}

int
main(int argc, char *argv[]) {
    pthread_t p1, p2;
    int rc;
    printf("main: begin\n");
    rc = pthread_create(&p1, NULL, mythread, "A"); assert(rc == 0);
    rc = pthread_create(&p2, NULL, mythread, "B"); assert(rc == 0);
    // join waits for the threads to finish
    rc = pthread_join(p1, NULL); assert(rc == 0);
    rc = pthread_join(p2, NULL); assert(rc == 0);
    printf("main: end\n");
    return 0;
}
```

- Threads share an address space; processes mostly do not.
- A critical section is any code that must not interleave with conflicting access.
- Locks protect invariants; races happen when the wrong interleaving matters.
- Know when a bug needs a lock versus a different synchronization tool.

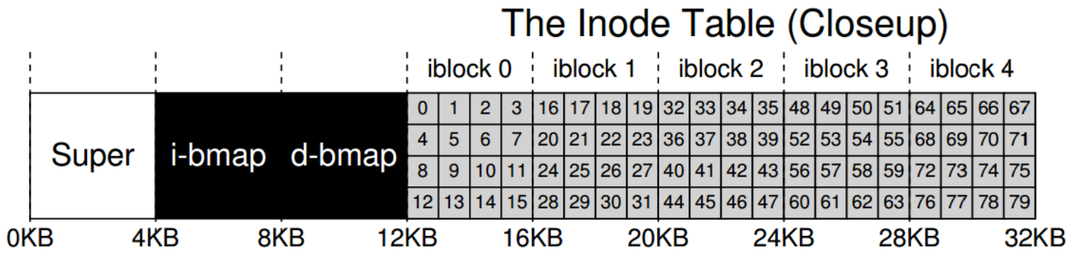
Condition variables and semaphores

- Condition variables wait for a predicate to become true.
- `pthread_cond_wait()` releases the lock and sleeps atomically.
- Use a `while` loop to re-check the condition after waking.
- Semaphores are useful for counting resources and handoff-style coordination.
- If you can explain the shared state and the predicate, you can usually choose the right primitive.

Storage and Filesystems

How the machine turns bytes into names, metadata, and blocks

Devices, disks, and files



- Devices are driven by controllers, drivers, interrupts, DMA, and sometimes polling I/O.
- HDDs care about seek time, rotational delay, and transfer time.
- Files, directories, links, and mount points are just the visible surface.
- Underneath, a file system is a giant data structure built from inodes, bitmaps, free-space structures, and block mappings.

File-system implementation

What operation happens between these two states?

```
inode bitmap  1110
inodes        [d a:0 r:3] [d a:1 r:3] [d a:2 r:2] []
data bitmap   1110
data          [(.,0) (..,0) (u,1)] [(.,1) (..,0) (k,2)] [(.,2) (..,1)] []
```

Command: [Select]

```
inode bitmap  1111
inodes        [d a:0 r:3] [d a:1 r:3] [d a:2 r:3] [d a:3 r:2]
data bitmap   1111
data          [(.,0) (..,0) (u,1)] [(.,1) (..,0) (k,2)] [(.,2) (..,1) (s,3)]
[(.,3) (..,2)]
```

- Be able to reason about what changes when a file is created, removed, linked, or written.
- Metadata updates matter just as much as the data itself.
- Be ready to trace inode and block-pointer changes.
- If you can trace the before/after state, you can usually solve the question.

Languages

How source text becomes something the machine can run

Grammars, lexers, and parsers

```
expr ::= expr + term | term
term ::= term * factor | factor
factor ::= ( expr ) | id | num
```

- BNF describes structure; the lexer turns characters into tokens.
- Predictive parsing uses grammar structure and lookahead to choose productions.
- Know the difference between **terminals**, **non-terminals**, and **productions**.
- If you can sketch an unambiguous parse tree, you parse the grammar.

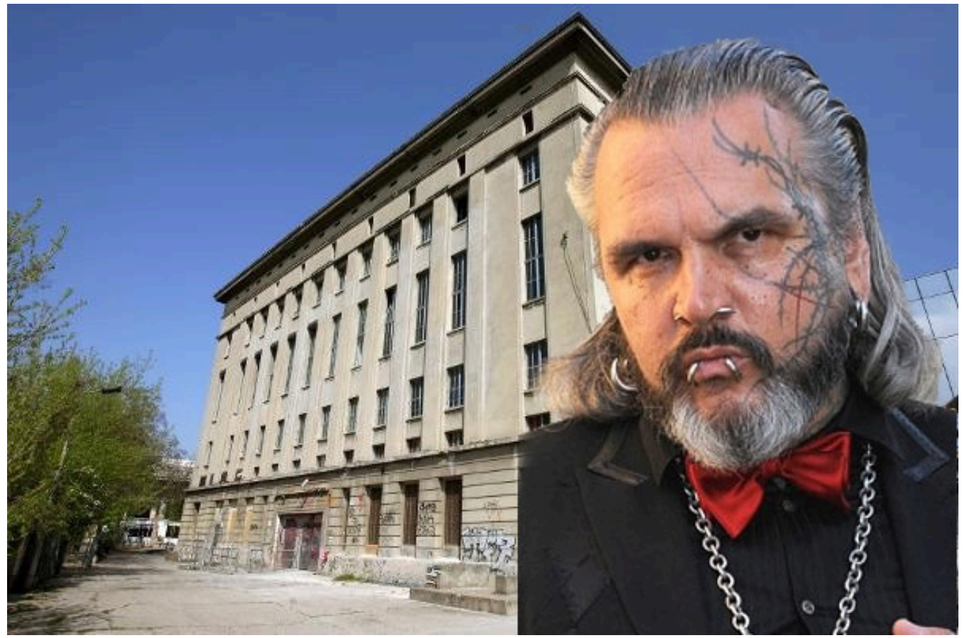
Compilers and interpreters

- A compiler **emits** code or another lower-level representation.
- An interpreter **evaluates** the program directly.
- The front end usually goes: source text -> tokens -> parse tree / AST -> emit or eval.
- Know where `emit()` and `eval()` happen and what each one receives.

Security

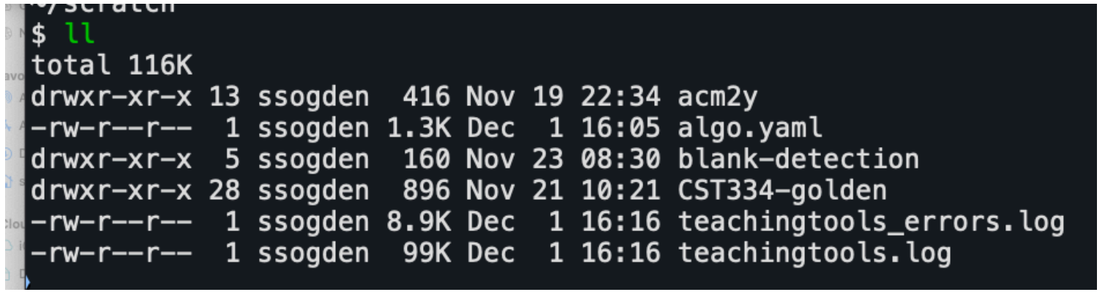
Who are you, what can you do,
and how do we keep that safe?

Authentication and access control



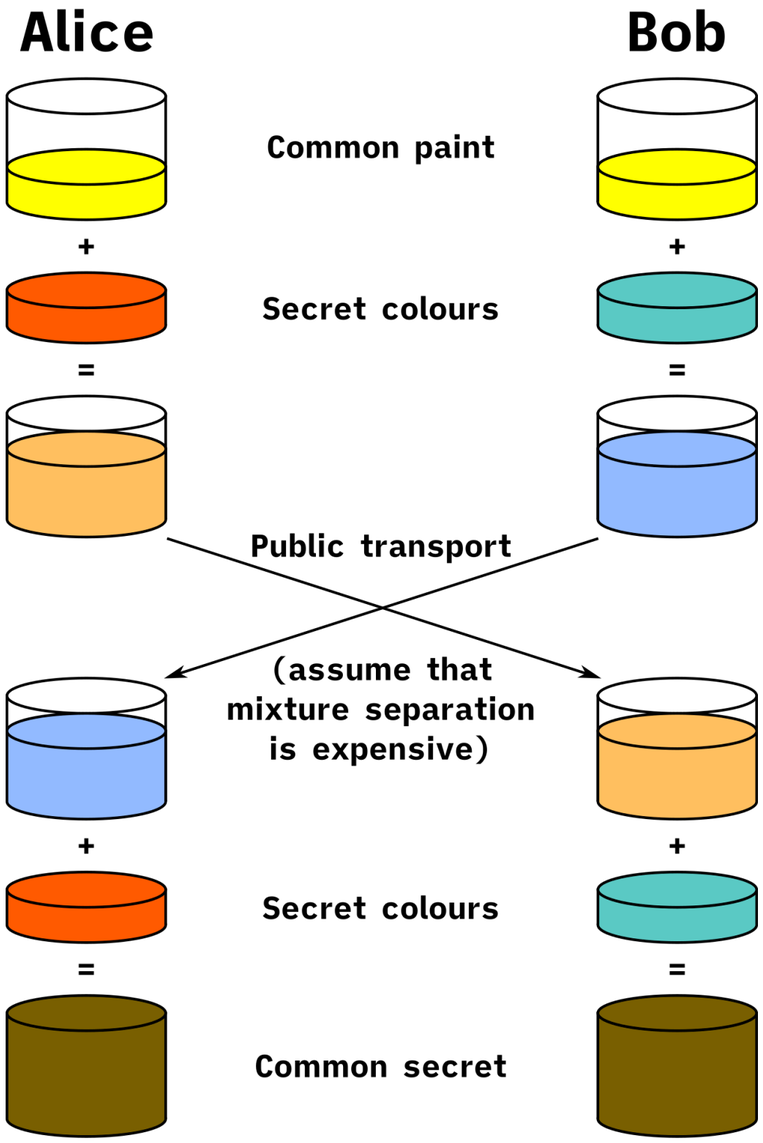
- **Authentication** answers “who are you?”
- **Authorization** answers “are you allowed?”
- Know the common **authentication** factors, salts, and why password hashes need help.
- Password hashing is about slowing attackers down, not making secrets magically safe.
- **ACLs**, **capabilities**, **RBAC**, and setuid-style examples are the big access-control mental models.

Access control



- **Access control** is about deciding whether a **subject** can touch an **object**.
- **ACLs** attach permission lists to the resource itself.
- **Capabilities** attach authority to the process or subject.
- **RBAC** and setuid-style examples show how real systems combine models.

Cryptography



- **Hashes**, **signatures**, **symmetric crypto**, and **public-key crypto** solve different problems.
- **Key exchange** is about getting a shared secret safely; trust is the hard part.
- Common failure modes are using the wrong primitive or using a primitive incorrectly.
- Be able to say what problem a crypto tool is actually solving.

Final checklist

- Can you explain each **mechanism** in your own words?
- Can you compare two **policies** and say what each one costs?
- Can you identify what state the OS or runtime is protecting?
- Can you say why the “obvious” answer is not always the right one?